

mizan — the balance ميزان

Computed citation verification for Qur'an & hadith answers

- Track 04: Knowledge & verification tools (+ Track 01 Q&A)
- The badge you see was computed, not asserted.
- bun run demo → VERIFIED + REJECTED in ~1s, offline, no key
- Islamic AI Challenge 2026 — islamicaich.org

1 The problem

LLMs hallucinate Islamic text with confidence

- • Western models dominate; Arabic religious data is scarce and unverified
- • A fluent fabrication looks identical to a real quote to a non-specialist
- • Measured: LLM hadith-verification accuracy $\approx 8\%$ (2026 study)
- • Fuzzy / embedding verifiers score invented-but-plausible hadith HIGH — CWE-345
- • Cost of error is not UX: false religious authority for worship & belief

2 The solution

Answer, then prove every quotation — per claim

- • [VERIFIED] quoted span literally contained in the cited record — nothing else
- • [REJECTED] cited record exists but does not contain the quote, and no anchor locates
- • [UNVERIFIABLE] no citation / empty quote / unresolved ID / timeout / anchor located
- • Every badge ships with evidence: quoted text + source row + source URL
- • Fail closed: zero evidence blocks approval; timeout → unverifiable; mismatch → no verdict

3 How it works

Six-step procedure · one Arabic fold · one route to VERIFIED

- 1 quote? 2 citation? 3 cap (≤ 3)? 4 resolve ID? 5 strict containment? 6 evidence?
- Fold table (one module): alef/hamza, wasla, ta-marbuta, diacritics, tatweel,
- Arabic-Indic digits → ASCII, punctuation/whitespace. Never adds/removes a letter.
- textMatch = matching key (never displayed) · textDisplay = what the user sees
- Verifier: 1 dependency (@mizan/core), no network, no clock, no randomness, no LLM judge

4 Why strict — the key innovation

No similarity path to VERIFIED. Proven, not promised.

- • Feasibility spike: invented hadith $\approx$ high fuzzy match $\rightarrow$ similarity IS the hole
- • So: no embedding, no edit-distance, no threshold, no percentage to tune
- • MatchStrength = exact|100 or none — a fuzzy 97% lives only in a display diagnostic
- • Gates enforce it: G-1 dependency closure · G-6 single VERIFIED site · verify.ts
- cannot import diagnostics — CI fails if a second route appears

5 Corpus & provenance

4 Sunan + Muwatta + Qur'an, 27,234 records · every number re-runnable

- • Qur'an Tanzil Uthmani 6,236 (verbatim, no-derivatives) + 4 Sunan + Muwatta
- • Grades never invented: stored as dataset asserts + source + basis, else null
- • 15,026 / 36,024 hadith rows quarantined (no grade) — served 20,998 + 6,236 verses
- • No Bukhari / Muslim — recorded as ABSENT, never substituted
- • attestation.json + hash-chained ledger · runs carry questionHash, never text

6 Evidence it works

200 golden · 40 red-team · demo in 60 seconds

- • golden-normalization 200/200: exact, undiacriticized, tatweel-spaced, digits must verify
- • red-team fabricated 40/40: one-word change, transpose, digit change → ZERO verified
- • Expectations hand-adjudicated in scripts/eval/plan.ts — generator cannot import verifier
- • bun install → bun run demo → bun run ci (typecheck + tests + gates G-1..G-7)
- • Benchmark vs plain FTS5 search re-runnable: bun run benchmark:vs-search

7 Honest limits

What we do NOT claim

- • No Sahih al-Bukhari / Sahih Muslim yet — biggest recall gap, stated upfront
- • Paraphrase: 26 elisions ruled UNVERIFIABLE via human-drawn anchors, not similarity
- Cost disclosed: all 40 fabrications move to UNVERIFIABLE; false verified stays 0
- • Default checkout = labelled PRECOMPUTED replay; live = hosted HTTPS allowlisted API
 - CLI + server-rendered demo forms (no client JS) · no tafsir backend (returns unavailable, never fabricated)
- • Golden bar is vs. fold-table behaviour, not vs. human-labelled corpus

8 Track fit & impact

Built for Track 04, usable from Track 01

- • Track 04: scholars & du'at get instant takhrij-check + grading provenance + referral
- • Track 01: any conversational answer inherits badges; non-verified → refer_to_scholar
- • Never a fatwa: verdict = containment fact, not truth/authenticity ruling
- • Impact: kill fabricated quotes at the source; make trust auditable in 60 seconds
- • Tech: Bun monorepo + TypeScript + Effect Schema decode + SQLite FTS5 + OpenAI-compatible LLM port

9 Reproduce it now

One command. No key. No network.

- • bun install --frozen-lockfile → bun run demo
- • Hosted demo: bun run demo-server → server-rendered verify playground, badges computed live
- • Full: bun run ingest → bun run ask "..." → bun run ci
- • Deliverables: public GitHub repo + hosted live demo + 2-min video + runbook + sources/licence registry
- • Synthetic data only · no PII · no secrets (gitleaks G-4) · Apache-2.0 code
- • Contact + repo QR here. نعلم الآلة.. لتخدم الرسالة — with proof.

10 Continuation plan

Hosting, roles, fallbacks, adoption, milestones

- • Hosting: Render free web service via Dockerfile + render.yaml — \$0 on the free tier;
- a model API is optional and operator-funded when a key is configured
- • Dependency fallback: FTS5 lexical baseline (bun run benchmark:vs-search) —
- retrieval can degrade; the verifier never can
- • Roles: maintainers own corpus ingest + gates; users file issues on the public repo;
- no private knowledge required to operate
- • Adoption: du'at & institutions use the CLI today; an MCP tool meets chat surfaces
- • Milestones: Bukhari & Muslim planned as graded ingest (not held today),
- Arabic UI, tafsir backend kept unavailable until a licensed source exists